

Laboratory Activity #5 — Packet Tracing (Detailed Step-by-Step)

Topic: Network Protocols & Communication — Packet Tracer lab

What you will build: PC1 → Switch → Router → Switch → PC2

Deliverables: topology screenshot, Simulation Mode screenshot showing packet path, completed protocol table, 3–5 sentence reflection.

Preparation (Before you start)

1. Open **Cisco Packet Tracer** on your computer.
2. Create a new file: **File → New**. Save early: **File → Save As** and name it PT_PacketTracing_Lastname.pt.

Step 1 — Place devices on the workspace

1. From the lower device panel select **End Devices → PC** and drag **2 PCs** onto the workspace. Rename them:
 - Right-click PC → **Rename** → PC1 and PC2.
2. From **Switches**, drag a **2960 Switch** onto the workspace. (You can use one switch for both PCs in a simple topology.)
3. From **Routers**, drag a basic router (e.g., **Router 1941**) onto the workspace.

Step 2 — Cable the devices

1. Click the **Connections** icon.
2. Choose **Copper Straight-through** for PC ↔ Switch and Switch ↔ Router links.
3. Connect:
 - PC1 FastEthernet0 → Switch FastEthernet0/1
 - PC2 FastEthernet0 → Switch FastEthernet0/2
 - Switch FastEthernet0/3 → Router GigabitEthernet0/0 (or appropriate router interface)
4. Verify each connection shows a green link light (click device → Physical to see NIC status if needed).

Step 3 — Configure Router interface (GUI & CLI)

GUI (recommended for beginners)

1. Click the **Router → CLI** tab. If prompted: press **Enter** to get to the prompt.

Commands to enter (type, then press Enter):

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# interface gigabitEthernet0/0
```

```
Router(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router# write memory
```

If your router uses fastEthernet0/0 instead of gigabitEthernet0/0, use that interface name.

Step 4 — Configure PC IP settings (GUI)

1. Click PC1 → **Desktop** → **IP Configuration**.
 - **IP Address:** 192.168.10.10
 - **Subnet Mask:** 255.255.255.0 (auto fills)
 - **Default Gateway:** 192.168.10.1
2. Repeat for PC2 with:
 - **IP Address:** 192.168.10.20
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.10.1

Step 5 — Quick connectivity check (Real Time)

1. On PC1 → **Desktop** → **Command Prompt**. Type:
ping 192.168.10.20
2. Expected: Reply from 192.168.10.20: bytes=32 time<1ms TTL=...
 - If replies are successful, good — move to simulation.
 - If Request timed out or Destination host unreachable, see **Troubleshooting** at the end.

Step 6 — Switch to Simulation Mode

1. Click the **Realtime/Simulation** toggle at bottom right and select **Simulation**.
2. In the Simulation panel (upper-right), set **Event List Filters** to show ICMP, ARP, IP, Ethernet (enable all these for visibility).
3. At the top toolbar, click **Add Simple PDU** (envelope icon) — or: **PDU** → **Create PDU**.

Step 7 — Create and send the PDU

1. Click PC1 (sender), then click PC2 (receiver). This creates a PDU from PC1→PC2.
2. In the Simulation control area press **Play Forward (Step)** once or press **Play** to run continuously.
3. Watch the animation: packets will move hop by hop. The Event List at the bottom shows each frame/packet with protocol and step.

Step 8 — Inspect packet details (deep dive)

1. While the packet is selected in the Event List, click **Event List** → **select a specific event**.
2. Click **Show/Inspect PDU** (or double-click the event) to open the PDU Details window.
You will see layered headers for:
 - **Ethernet II** (MAC addresses)
 - **Internet Protocol (IP)** (source & destination IPs)
 - **Internet Control Message Protocol (ICMP)** (type = Echo Request/Reply)
3. Expand each header to read fields like **Source MAC**, **Destination MAC**, **IP TTL**, **ICMP Type**, etc.

Step 9 — Observe ARP behavior

1. If the destination MAC is unknown, Packet Tracer will show an **ARP Request** from PC1 with broadcast MAC ff:ff:ff:ff:ff:ff.
2. Inspect the ARP event in the event list — you'll see the ARP reply with the MAC address of PC2. Note which device answered the ARP (PC2) and where the router participates (if inter-networking).

Step 10 — Document your observations (complete the table)

For at least three events (ARP request, ICMP Echo Request, ICMP Echo Reply), fill the table:

Layer/Step	Device	Protocol Observed	What changed in headers (example)
e.g., 1	PC1	ARP	Broadcast ARP, source IP 192.168.10.10, dest MAC ff:ff...
2	Switch	Ethernet	Frame forwarded; switch looked up MAC and forwarded to correct port
3	Router	IP	Packet inspected, TTL decremented (if routed)
4	PC2	ICMP	Echo Reply created, source/dest IP swapped

Step 11 — Capture screenshots

1. **Topology screenshot:** Zoom so whole topology is visible → **PrtSc** or Packet Tracer **File** → **Export** → **PNG** (if available).
2. **Simulation screenshot:** Pause the simulation while a packet is in transit and capture the Packet Tracer window showing the packet path and Event List.

Step 12 — Save and submit

1. Save your Packet Tracer file: **File** → **Save**.
2. Submit via LMS: upload PT_PacketTracing_Lastname.pt, the two screenshots, the completed table, and the 3–5 sentence reflection.

Troubleshooting (common problems & fixes)

- **No ping / Destination host unreachable**
 - Verify PC IP/gateway matches router network. Recheck subnet mask.
 - Check router interface is not shutdown and has the correct IP.
 - Ensure cables are correct (straight-through for PC↔Switch).
- **Switch ports show red (down)**
 - Click device → Physical → ensure NIC is present and powered.
- **ARP not visible**
 - Make sure Event List Filters include ARP and that you started a fresh PDU (clear previous PDUs).
- **Packets stuck at router**
 - Check router interface IP configuration and interface status in CLI (show ip interface brief). In Packet Tracer CLI:
 - Router# show ip interface brief
 - Router# show running-config
- **Simulation shows broadcasts flooding switch**
 - That's expected for ARP broadcasts. Observe how only the necessary destination responds.